LTspice Exercise

Prepare a report (and submit in google classroom) briefly summarizing the observations (including IV and extracted parameters) and conclusions from the experiments below.

S1. Read the card.

For `L=65n` and `L=70n`, compute L_{eff} and W_{eff} from the card, and C_{ox} from `TOXE`. Then run `.op` at VGS = VDS = 1.1 V and note Id for both cases and comment on the observed trend.

S2. Transfer characteristic.

Simulate I_D–V_G at V_{DS} = 50 mV and 1.1 V (L = 65 nm). Export the data and use Python to plot both on linear and logarithmic axes. Extract V_{TH}, I_{ON} and I_{OFF}, and compare them with the table of Section 5.1.

S3. Output characteristic.

Produce the I_D–V_D family at V_{GS} = 0.3, 0.6, 0.9 and 1.5 V. Compute g_{ds} with `np.gradient` and report $r_o = 1/g_{ds}$ in saturation at the highest V_{GS}, in kΩ·µm.

S4. Physics check.

Re-run S2 once with `UA = UB = 0` and once with `VSAT` multiplied by 100. Overlay the three curves and explain which TCAD model each of these corresponds to (`Enormal`, `HighFieldSaturation`).

S5. Parametric Variation.

Simulate I_D–V_G at V_{DS} = 0.4 V and I_D–V_D at V_{GS} = 1.1 V for each of the following cases (experiments 1 to 13) where one parameter is varied around the nominal value to assess its impact. Import the data into Python and plot it neatly with legend showing which curve corresponds to what value of the parameter in each experiment. Explain the reason for the trend observed in each case by connecting it with device physics (e.g. trend in variation of V_{TH}, I_{ON} and I_{OFF}). In cases where the observation is not matching your device physics understanding explain why that could be because of the way SPICE treats that parameter.

Exp.	Parameter	Values (low / nominal / high)
1	VTH0	0.30 / **0.423** / 0.55 V
2	TOXE	1.5 nm / **1.85 nm** / 2.5 nm
3	L (drawn)	55 nm / **65 nm** / 100 nm
4	XL	−35 nm / **−30 nm** / −15 nm
5	LINT	2 nm / **5.25 nm** / 8 nm
6	W (drawn)	0.5 µm / **1 µm** / 2 µm
7	WINT	0 nm / **5 nm** / 40 nm
8	NDEP	1.5×10^{18} / **2.54×10^{18}** / 4×10^{18} cm^{-3}
9	NSD	5×10^{19} / **2×10^{20}** / 5×10^{20} cm^{-3}
10	XJ	10 nm / **19.6 nm** / 30 nm
11	NGATE	1×10^{19} / **2×10^{20}** / 1×10^{21} cm^{-3}
12	U0	300 / **491** / 700 cm²/V·s
13	VSAT	0.8×10^{7} / **1.24×10^{7}** / 2×10^{7} cm/s
14	RDSW	50 / **165** / 400 Ω·µm

S6. Approximate match.

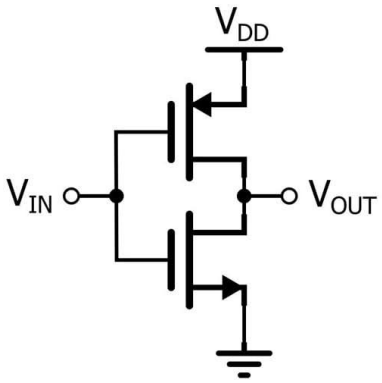
Following Module 6 in this handout and from your own understanding, tune `nfit` to match your TCAD curves: ID–VG at VDS = 0.4 V and ID–VD at VGS = 0.9 V, both on linear axes, as much as possible (perfect match may not be possible!). Deliver overlays of TCAD (markers) and SPICE (lines), and also a table of the parameters you changed —mention the PTM value, and your value.

S7. A sample circuit simulation.

Following is the schematic of a CMOS inverter. Simulate it choosing the below parameters (all others default)

Parameter	nMOS	pMOS
L	65 nm	65 nm
W	1 µm	2 µm
VDD	1.1 V	1.1 V

Obtain the Vout versus Vin plot (sweeping Vin from 0 to VDD) and from that explain why this is called a CMOS "*inverter*".



S8. GitHub.

All python codes used in the above experiments are to be pushed to your GitHub profile in a separate repository named LTSPICE. This shall be used during end-sem viva.